Class XII Maths Sample Paper – Set 1

Time Allowed: 30 Minutes

General Instructions:

1. This Question paper contains - five sections A, Each section is compulsory. However, there are internal choices in some questions.

2. Section A has 18 MCQ's and 02 Assertion-Reason based questions of 1 mark each.

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Section A		
Q. No.	**Questions**	**Marks**
1.	If $f(x) = x^2 + 4x - 5$ and $A = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$, then $f(A)$ is equal to:	1

	(a) $\begin{bmatrix} 8 & 4 \\ 4 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} 12 & 8 \\ 8 & 12 \end{bmatrix}$ (c) $\begin{bmatrix} -2 & 4 \\ 4 & -2 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & 2 \\ 2 & 0 \end{bmatrix}$	
2.	If A is a 3-rowed square matrix and $\lvert A \rvert = 4$, then $\text{adj}(A)$ is: (a) $128A$ (b) $64A$ (c) $4A$ (d) $16A$	1
3.	For any 2×2 matrix, if $A(\text{adj}A) = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$, then $\lvert A \rvert$ is equal to:	1

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	(a) 20 (b) 10 (c) 0 (d) 100	
4.	If $f(x) = \|x\| + \|x-2\|$, then: (a) $f(x)$ is continuous at $x=0$ but not at $x=2$ (b) $f(x)$ is continuous at $x=0$ and $x=2$ (c) $f(x)$ is continuous at $x=-2$ but not at $x=0$ (d) $f(x)$ is continuous at $x=2$ but not at $x=0$	1
5.	If a line makes angles $\frac{\pi}{3}, \frac{3\pi}{4}$ with the X-axis and Y-axis respectively, then the angle which it makes with the Z-axis is: (a) π b) $\frac{\pi}{2}$ c) 0° d) both 0° and π	1

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6.	Which of the following is the integrating factor of $(x\log x)\dfrac{dy}{dx}+y=2\log x$? a) $\dfrac{1}{x\log x}$ (b) $\dfrac{1}{\log x}$ (c) $x\log x$ (d) $x^2\log x$	1
7.	item Which of the following statements is correct? (a) Every LPP admits an optimal selection. (b) A LPP admits a unique optimal solution. (c) If a LPP admits two optimal solutions, it has an infinite solution. (d) The set of all feasible solutions of a LPP is not a convex set. (a) Option(d) (b) Option(a) (c) Option(b) (d) Option(c)	1
8.	$\bar{a}+\bar{b}+\bar{c}=0$ .such that $\lvert\bar{a}\rvert=3$, $\lvert\bar{b}\rvert=5$, and $\lvert\bar{c}\rvert=7$. What is the angle between $\bar{a}$ and $\bar{b}$?	1

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	(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{9}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$	
9.	$\int \frac{\sin x}{1+\cos x} dx = ?$ (a) $x + \tan x - \sec x + C$ (b) $x + \frac{2}{\tan\frac{x}{2}+1} + C$ (c) $x - \tan x - \sec x + C$ (d) $x - \tan x + \sec x + C$	1
10.	For what value of x , the matrix $A = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ x & -3 & 0 \end{bmatrix}$ is skew-symmetric?	1

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	(a) $x = 2$ (b) $x = -2$ (c) $x = 1$ (d) $x = 3$	
11.	The linear programming problem to minimize $Z = 3x + 2y$ subject to constraints $x + y \geq 8$, $3x + 5y \leq 15$, $x \geq 0$, and $y \geq 0$, has: (a) no feasible solution (b) one solution (c) infinitely many solutions (d) two solutions enditemize}	3
12.	item If $\bar{a} = (\hat{i} + 2\hat{j} - 3\hat{k})$ and $\bar{b} = (3\hat{i} - \hat{j} + 2\hat{k})$, then the angle between $(\bar{a} + \bar{b})$ and $(\bar{a} - \bar{b})$ is:	3

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	(a) $\frac{\pi}{4}$ (b) $\frac{2\pi}{3}$ (c) π (d) $\frac{\pi}{2}$	
13.	The existence of the unique solution of the system of equations: $x + y + z = \lambda$ $5x - y + \mu z = 10$ $2x + 3y - z = 6$ depends on a) λ and μ b) λ only c) neither both λ nor μ d) μ only	4
14.	In a certain town, 40% persons have brown hair, 25% have brown eyes, and 15% have both. If a person selected at random has brown hair, the chance that a person selected at random with brown hair is with brown eyes. a) 1/3 b) 3/20	4

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15.	The solution of the differential equation $xdx + ydy = x^2 ydy - y^2 xdx.$ is: (a) $x^3 + 1 = C(1 - y^3)$ (b) $x^3 - 1 = C(1 + y^3)$ c) $x^2 + 1 = C(1 - y^2)$ d) $x^2 - 1 = C(1 + y^2)$	4
16.	If $\bar{\beta}$ is perpendicular to both α and γ, where $\alpha = \hat{k}$ and $\gamma = 2\hat{i} + 3\hat{j} + 4\hat{k}$, then what is $\bar{\beta}$ equal to? a) $-2\hat{i} + 3\hat{j}$ b) $3\hat{i} + 2\hat{j}$ (c) $2\hat{i} - 3\hat{j}$ d) $-3\hat{i} + 2\hat{j}$	6
17.	If $y = \sin^{-1} x$, then $(1 - x^2) y_2$ is equal to: (a) xy (b) xy_1 (c) xy_2	6

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	(d) x^2.	
18.	A line passes through the point A(5, -2, 4) and it is parallel to the vector $(2\hat{i} - \hat{j} + 3\hat{k})$. The vector equation of the line is: (a) $\bar{r} = (5\hat{i} - 2\hat{j} + 4\hat{k}) = \sqrt{14}$ b) $\bar{r} = (5\hat{i} + 2\hat{j} - 4\hat{k}) = \sqrt{12}$ (c) $\bar{r} = (5\hat{i} - 2\hat{j} + 4\hat{k}) + \lambda(2\hat{i} - \hat{j} + 3\hat{k})$ d) $\bar{r} = (2\hat{i} - \hat{j} + 3\hat{k}) + \lambda(5\hat{i} - 2\hat{j} + 4\hat{k})$	1
19.	Assertion (A): If x is real, then the minimum value of x^2 - 8x + 17 is 1. Reason (R): If f''(x) > 0 at a critical point, then the value of the function at the critical point will be the minimum value of the function. a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A. c) A is true but R is false. d) A is false but R is true	1
20.	Assertion (A): If A=x∈z:0≤x≤12} and R is the relation in A given by R = (a, b) : a = b. Then, the set of all elements related to 1 is 1, 2}. Reason (R): If R1 and R2 are equivalence relation in a set A, then R2 ∩ R2 is an equivalence relation	1

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	a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A. c) A is true but R is false. d) A is false but R is true.	

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